ELSEVIER

Precambrian Research 69 (1994) 281–292

Precambrian Research

Palaeomagnetic and petrological reconstruction of the Seychelles

Kanenori Suwa[a], Katsuyasu Tokieda[b], Mitsuo Hoshino[c]

[a]*Laboratory of Earth Sciences, Nihon Fukushi University, Okuda, Mihama-chô, Chita-gun, Aichi 470-32, Japan*
[b]*Department of Physics, Faculty of Science, Shimane University, Nishikawazu-chô, Matsue 690, Japan*
[c]*Department of Natural Science Informaties, School of Informaties and Sciences, Nagoya University, Chikusa-ku, Nagoya 464-01, Japan*

Received September 1, 1992; revised version accepted January 27, 1994

Abstract

The geology of Mahé-Island and adjoining islets, Seychelles, consists of three types of late Precambrian granitic rock series: (1) a gneissose granodiorite–gneissose tonalite and amphibolite–leucogranodiorite series, (2) a porphyritic granite–pink granite series, and (3) a grey granite–gneissose grey granite series. All these granitic rocks are cut by dykes of doleritic compositions, mostly with a WNW–ESE trend. The gneissose granodiorite is a calc-alkaline I-type granite with normative Ab ≫ Or, the porphyritic granite is a sub-alkaline A-type granite with normative Ab > Or, and the grey granite is an alkaline A-type granite with normative Ab > Or. Rb–Sr radiometric analyses yielded ages of 713 ± 19 Ma for the gneissose granodiorite, 683 ± 16 Ma for the porphyritic granite and 570 ± 5 Ma for the grey granite.

Petrological, geological, geochronological and geochemical studies suggest that the late Precambrian granitic rocks of the Seychelles are very similar to, and are well correlated with those of the Arabian–Nubian shield.

408 late Precambrian granite specimens were collected from the Mahé island group, Seychelles. Progressive alternating field demagnetization and thermal demagnetization have been successfully performed to isolate stable remanence with well-grouped direction from natural remanent magnetization of the granites. The stable remanences of the granite fall around a northerly mean direction, of which the declination is 348.5° and the inclination +48.7°. The stable remanences of the Mahé island group granite yield a palaeomagnetic pole at 54.0°N, 38.2°E.

From the above data, in conjunction with reported polar wander path and plate rotation poles, it is concluded that the Seychelles were originally located near the eastern end of the Horn of Africa and have drifted to the present position since continental break-up.

1. Introduction

The Mahé–Praslin islands, the principal group of the Seychelles Bank situated 1,700 km east of Mombasa, are composed largely of Precambrian granitic rocks. The Mahé–Praslin group rises from the central part of the submarine Seychelles Bank which is less than 60 m below sea-level and covers 41,000 km^2. The Seychelles Bank is the northern extremity of the Mascarene Plateau linking the Seychelles, Réunion and Mauritius (Fig. 1) and being free from earthquake activity, and can well be called a "microcontinent".

According to seismic refraction experiments (Davies and Francis, 1964; Francis et al., 1966; Matthews and Davies, 1966; Laughton et al., 1970), five layers have been recognized on the Seychelles Bank: (1) a superficial layer, (2) a 5.72 km/s layer, (3) a 6.26 km/s layer, (4) a 6.78 km/s layer, and (5) a 8.1 km/s layer. Ve-

0301-9268/94/$07.00 © 1994 Elsevier Science B.V. All rights reserved
SSDI 0301-9268(94)00027-O

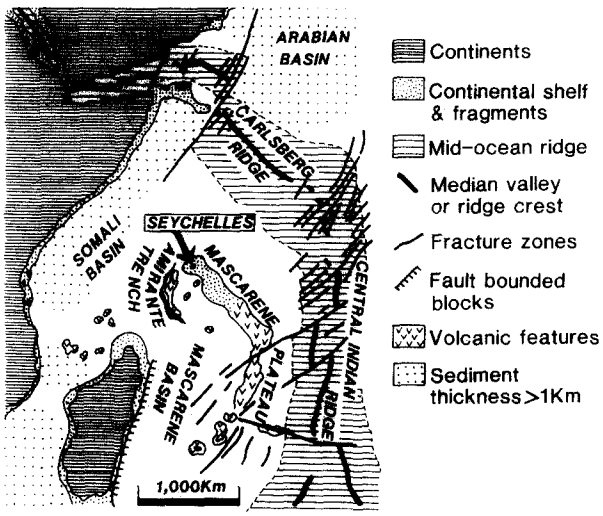


Fig. 1. Geological situation of the Seychelles (after Laughton et al., 1970).

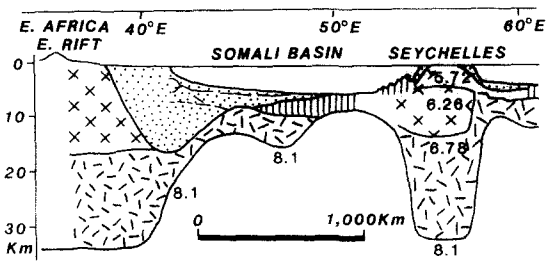


Fig. 2. Underground structure of the Seychelles, Somali Basin, and East Africa (after Laughton et al., 1970).

locities of 5.72 and 6.26 km/s are typical of granitic rocks and the interface between the 6.78 km/s layer and the 8.1 km/s layer at 33 km is interpreted as the Mohorovičić discontinuity (Fig. 2). The granitic rocks extend to a depth of 15 km and the crust there has both a continental character and a continental thickness.

The first detailed geological survey of the Seychelles Archipelago was made by Baker (1963). He recognized three main types of Precambrian granite: i.e., (1) a grey Mahé granite, (2) a grey faintly gneissose granite, and (3) a reddish Praslin–La Digue granite. A swarm of dolerite dykes is found cutting all three types of granite. Early Tertiary syenite and diorite occur in the Silhouette and North islands. Petrographic and petrochemical studies on the rocks described above have been reported by Baker (1963), Frankel and Kent (1964), Michot and Deutsch (1977), Agata and Suwa (1983), Suwa et al. (1983), Hoshino (1986) and Richter (1987). Agata and Suwa (1983) reported mineralogical data on magnetite and ilmenite and discussed how the MnO contents in ilmenite vary with differentiation in host granitic magma. Richter (1987) provided the data on trace and rare earth element abundances in the granites.

Radiometric dating was performed on whole-rock samples and rock-forming minerals from the Seychelles by Miller and Mudie (1961), Baker and Miller (1963), Wasserburg et al. (1963), Yanagi et al. (1983), Weis and Deutsch (1984) and Hargraves and Duncan (1990). Their results showed that the emplacement of the Seychelles granitic rocks occurred in late Precambrian time, whereas the syenite and diorite exposed on the Silhouette and North islands are early Tertiary in age. The emplacement of dolerite dykes occurred twice, once in the late Precambrian and once in the early Tertiary.

Palaeomagnetic studies of the dolerite dykes by Matthews and Reilly (1964) were unsuccessful. Palaeomagnetic studies were reported by Tokieda et al. (1983), who found that the stable palaeomagnetic remanence of the Precambrian granitic rocks yielded to a northerly mean direction with a declination of 348.5° and an inclination of +48.7° ($k=39$ and $\alpha_{95}=4.8°$). Hargraves and Duncan (1990) performed successfully their palaeomagnetic studies on the dolerite dykes and proposed the reconstruction model for the Seychelles.

The Seychelles Bank is believed to be continental fragments left behind during the disruption of Gondwanaland and the opening of the Indian Ocean. A relative plate motion scenario for this history is provided by Emerick and Duncan (1982) and Hargraves and Duncan (1990).

The object of this paper is to propose a reconstruction model of the Seychelles from further palaeomagnetic and petrological studies of the Precambrian granitic rocks on Mahé and the adjoining islets.

2. Geological setting

Suwa et al. (1983) subdivided the Precambrian granitic rocks of Mahé Island into three types, namely (1) a gneissose granodiorite–gneissose tonalite and amphibolite–leucogranodiorite series, (2) a porphyritic granite–pink granite series, and (3) a grey granite–gneissose grey granite series as shown in Fig. 3. All these granitic rocks are cut by dykes of doleritic compositions, mostly with a WNW–ESE trend.

2.1. Gneissose granodiorite–gneissose tonalite and amphibolite–leucogranodiorite series

This rock series occurs mainly in the northern part of Mahé Island. The radiometric age of the gneissose granodiorite is 713 ± 19 Ma (Yanagi et al., 1983).

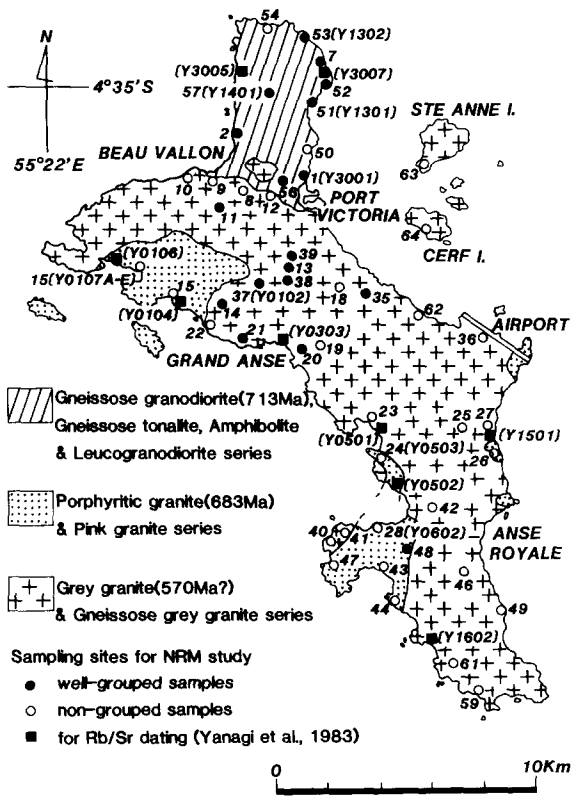


Fig. 3. Geological map of the Mahé island group, Seychelles (modified from Suwa et al., 1983), with sampling sites for palaeomagnetic and geochronological studies.

The gneissose granodiorite is heterogeneous with many ellipsoidal and angular mafic enclaves. The gneissose tonalite occurs near the boundary between the gneissose granodiorite and amphibolite, and is considered to be a result of contamination of amphibolite by gneissose granodiorite. These rocks have parallel banding and foliation, though these are faint. They are coarse-grained in some places and fine-grained in other places. The ellipsoidal and angular mafic enclaves, 5–20 cm across, are arranged parallel to the foliation. The foliation has a NNE trend and a steep dip to the southeast.

The leucogranodiorite is generally homogeneous and becomes heterogeneous near the boundary between the leucogranodiorite and the gneissose granodiorite–gneissose tonalite and amphibolite. The latter three are penetrated discordantly by the leucogranodiorite and are agmatitic in appearance.

2.2. Porphyritic granite–pink granite series

Porphyritic granite occurs along the west coast of Mahé Island. The radiometric age of the porphyritic granite is 683 ± 16 Ma (Yanagi et al., 1983). Pink granite occurs sporadically as small bodies along the east coast and on the southwest coast of Mahé Island.

The porphyritic granite is characterized by its texture with phenocrysts of feldspar. The pink granite, salmon-pink to brownish pink in colour, is coarse-grained equigranular plutonic rock and is similar to porphyritic granite in petrography with exception of its non-porphyritic texture.

2.3. Grey granite–gneissose grey granite series

Grey granite occurs in the main part of Mahé Island and gneissose grey granite occurs in Cerf Island and Ste Anne Island. The apparent isochron age of the grey granite is 570 ± 5 Ma (Yanagi et al., 1983).

The grey granite is coarse-grained equigranular plutonic rock and is massive with no foliation. Mafic enclaves are rather small and few in this rock. These petrographic characters are uniformly constant throughout the body. The grey

granite body in a stock-like form is found to the northwest of Port Victoria and this body seems to have been emplaced discordantly in the gneissose granodiorite.

The gneissose grey granite petrographically resembles the grey granite. The nature of this faintly gneissose structure suggests that it is due to primary flow banding. Petrographically the gneissose grey granite is quite distinct from the gneissose granodiorite.

2.4. Mutual relationships between the three granite series

From the field relations, it is apparent that the grey granite is younger than the gneissose granodiorite and even younger than the leucogranodiorite. Although the field relations between porphyritic granite, pink granite and grey granite appear to be of a gradual character, Rb/Sr isotopic ratios suggest that the grey granite is the most evolved rock among them (Yanagi et al., 1983). Therefore the grey granite is thought to be the latest intrusion on Mahé Island.

Thus, the major intrusive events which took place in Mahé are conceived as follows. First event: gneissose granodiorite at about 713 Ma ago, and followed by leucogranodiorite. Second event: porphyritic granite and pink granite at about 683 Ma ago. Third event: grey granite with a marginal gneissose grey granite at about 570 Ma ago.

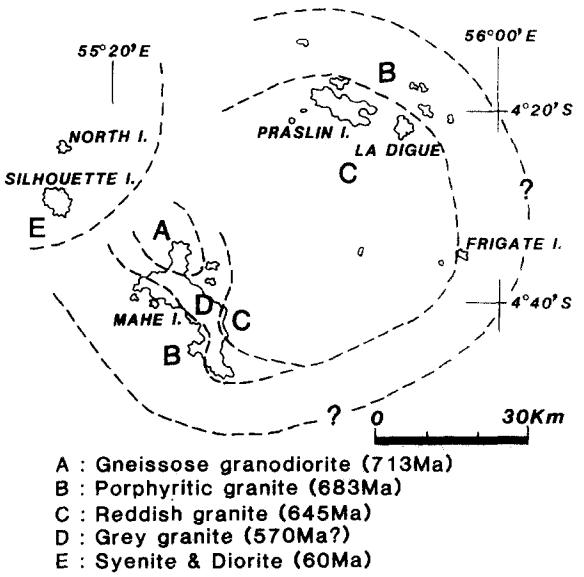


Fig. 4. Geological sketch map of the Mahé–Praslin island group, Seychelles (revised from Suwa et al., 1983 and Suwa, 1988).

Fig. 4. shows the geological sketch map of the Mahé–Praslin group. The gneissose granodiorite (A) is mainly distributed in the north of Mahé Island. The porphyritic granite (B) is considered to encircle the main mass of the body. The major part of the Mahé–Praslin group of islands is occupied by the reddish granite (C) and grey granite (D). Syenitic and dioritic rocks (E) of early Tertiary age (60 Ma) are distributed around the Silhouette and North islands and these young intrusives occupy the northwestern margin of the Mahé–Praslin group.

3. Petrography of Precambrian granitic rocks

3.1. Gneissose granodiorite–gneissose tonalite and amphibolite–leucogranodiorite series

The gneissose granodiorite shows a medium- to coarse-grained granoblastic texture with occasional porphyritic feldspars. This rock consists mainly of oligoclase, quartz, microcline–microperthite, biotite and hornblende in a descending order. Accessories are sphene, allanite, apatite, zircon, magnetite and ilmenite. Small amounts of ferroaugite and ferroactinolite occur enclosed in hornblende. TiO_2 content in the magnetite is less than 0.02 wt.%. MnO contents in the ilmenite intergrown with magnetite and in discrete ilmenite are up to 19 wt.% and 11 wt.%, respectively.

The leucogranodiorite shows a medium- to coarse-grained granoblastic texture. This rock consists mainly of oligoclase, quartz, microcline–microperthite and biotite with accessories of hornblende, sphene, apatite, zircon, allanite, magnetite and ilmenite. TiO_2 content in the magnetite is less than 0.46 wt.%. MnO contents in the ilmenite intergrown with magnetite and in discrete ilmenite are up to 32 wt.% and 26 wt.%, respectively.

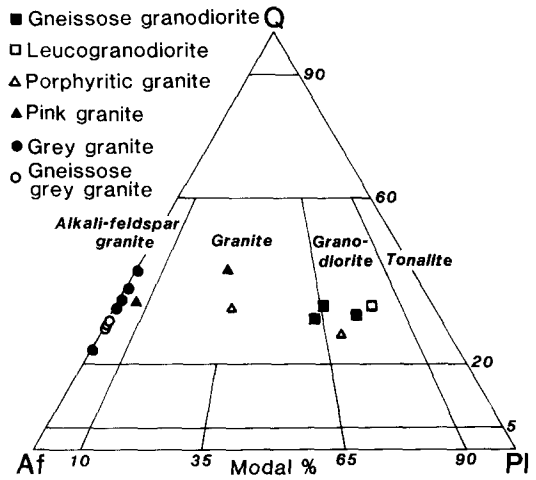


Fig. 5. Modal compositions of the granitic rocks of the Seychelles.

3.2. Porphyritic granite–pink granite series

The porphyritic granite consists of albite, quartz, microcline–microperthite, ferroedenite and biotite with accessories of fluorite, sphene, apatite, zircon, allanite, magnetite and ilmenite. Phenocrystic microperthite is commonly rimmed with albite. Stilpnomelane occurs as subsolidus minerals. TiO_2 content in the magnetite is less than 0.45 wt.%. MnO content in ilmenite intergrown with magnetite is 6 wt.%.

The pink granite shows a coarse-grained granoblastic texture and this rock consists of microcline–mesoperthite, quartz, albite, ferroedenite and biotite with accessories of fluorite, sphene, apatite, zircon, allanite, magnetite and ilmenite. Subsolidus stilpnomelane flakes commonly occur fringing the primary ferromagnesian minerals. TiO_2 content in the magnetite is less than 0.33 wt.%. MnO content in discrete ilmenite is 11 wt.%.

3.3. Grey granite–gneissose grey granite series

The grey granite is coarse-grained and exhibits a hypidiomorphic equigranular texture. This rock consists of microcline–mesoperthite, quartz, ferrorichterite–ferrowinchite, ferroactinolite, riebeckite, ferrohedenbergite and annite with accessories of albite, fluorite, sphene, apatite, zircon, allanite, magnetite and ilmenite. Albite seams can be seen between the mesoperthite grains. Two kinds of subsolidus stilpnomelane (a reddish-brown ferristilpnomelane and a green ferrostilpnomelane) are observed. TiO_2 content in the magnetite is less than 0.75 wt.%. MnO contents in ilmenite intergrown with magnetite and in discrete ilmenite are 4 wt.% and 6 wt.%, respectively.

The gneissose grey granite is medium-grained and shows a xenomorphic equigranular texture. The mineral assemblage of this rock is quite similar to the grey granite. Opaque mineral occupies only 0.1–0.2 % in volume. MnO content in the ilmenite intergrown with magnetite (now replaced by haematite) is only 1.6 wt.%.

Fig. 5 shows the modal compositions of the Precambrian granitic rocks of the Seychelles.

4. Petrochemistry and geochronology of Precambrian granitic rocks

Chemical compositions and C.I.P.W. Norms of the Precambrian granitic rocks in the Mahé island group, the Seychelles, are shown in Table 1. The differentiation indices (normative Q+Or+Ab) of the gneissose granodiorite, porphyritic granite and pink granite, and grey granite and gneissose grey granite are 85.8, 93.0, and 94.1, respectively. Normative albite is more than normative orthoclase; it is especially clear in the gneissose granodiorite, Ab (44.3) $\gg$ Or (14.2), and it is clear in the gneissose grey granite, Ab (34.6) > Or (28.5).

CaO contents are exceedingly low in the grey granite and gneissose grey granite. The $(Na_2O+K_2O)/Al_2O_3$ molecular ratio varies: 0.74 for the gneissose granodiorite, 0.93 for the porphyritic granite, 0.94 for the pink granite, 0.93 for the grey granite, and 1.03 for the gneissose grey granite (Fig. 6). Iron contents are fairly high in all types of rock. From these chemical properties and mineral parageneses, the gneissose granodiorite is defined as calc-alkaline granite, the porphyritic granite and the pink granite as sub-alkaline granite, and the grey granite and the

Table 1
Chemical compositions and C.I.P.W. norms of the granitic rocks of the Seychelles

	1 Gneissose granodiorite aver. (4)	2 Gneissose tonalite and amphibolite aver. (3)	3 Leuco-granodiorite aver. (2)	4 Porphyritic granite aver. (2)	5 Pink granite	6 Grey granite	7 Gneissose grey granite aver. (2)	8 Average composition of granitic rocks (1, 3, 4, 5, 6, 7)
SiO_2	72.25	48.62	74.93	74.00	73.54	75.43	75.50	74.28
TiO_2	0.34	1.27	0.24	0.30	0.36	0.22	0.24	0.28
Al_2O_3	15.12	18.80	13.20	12.94	13.16	12.13	11.96	13.09
Fe_2O_3	1.24	4.05	1.17	0.81	1.30	0.60	0.83	0.99
FeO	1.10	5.77	0.69	1.57	1.21	1.67	1.16	1.23
MnO	0.073	0.18	0.063	0.078	0.097	0.087	0.029	0.071
MgO	0.52	4.65	0.23	0.43	0.41	0.17	0.30	0.34
CaO	2.08	9.29	0.77	1.11	0.88	0.48	0.27	0.93
Na_2O	5.23	3.63	4.99	4.72	4.72	4.60	4.33	4.77
K_2O	2.41	1.46	3.98	3.89	4.93	4.06	4.83	4.02
P_2O_5	0.075	0.24	0.033	0.053	0.092	0.024	0.029	0.051
H_2O^+	0.38	0.96	0.24	0.36	0.18	0.27	0.21	0.27
H_2O^-	0.13	0.67	0.14	0.11	0.11	0.23	0.29	0.17
Total	100.95	99.59	100.68	100.37	100.99	99.97	99.98	100.49
Q	27.33	0.00	29.14	28.48	25.42	31.22	30.86	28.68
C	0.31	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Or	14.24	8.63	23.52	22.99	29.14	23.99	28.54	23.76
Ab	44.26	29.51	42.23	39.94	39.94	38.93	34.63	40.36
An	9.83	30.69	1.86	2.63	0.16	0.46	0.00	2.43
Ne	0.00	0.65	0.00	0.00	0.00	0.00	0.00	0.00
Ac	0.00	0.00	0.00	0.00	0.00	0.00	1.77	0.00
En (Di)	0.00	3.67	0.57	0.39	0.84	0.12	0.16	0.33
Es (Di)	0.00	1.74	0.02	0.68	0.60	0.68	0.34	0.44
Wo (Di)	0.00	5.77	0.68	1.05	1.50	0.74	0.48	0.77
Wo	0.00	0.00	0.04	0.00	0.00	0.00	0.00	0.00
En (Hy)	1.28	0.00	0.00	0.68	0.18	0.30	0.59	0.51
Fs (Hy)	0.57	0.00	0.00	1.18	0.13	1.69	1.27	0.67
Fo (Ol)	0.00	5.55	0.00	0.00	0.00	0.00	0.00	0.00
Fa (Ol)	0.00	2.90	0.00	0.00	0.00	0.00	0.00	0.00
Mt	1.80	5.87	1.70	1.17	1.88	0.87	0.32	1.44
Il	0.65	2.41	0.46	0.57	0.68	0.42	0.46	0.53
Ap	0.17	0.56	0.08	0.12	0.21	0.06	0.07	0.12

(Analyst: R. Sugisaki).

gneissose grey granite as alkaline granite. These chemical properties also show that the gneissose granodiorite belongs to the I-type granite, and that the porphyritic granite, the pink granite, the grey granite and the gneissose grey granite belong to the A-type granite.

Rb–Sr radiometric dating was performed on whole-rock samples of granitic rocks from the Seychelles (Yanagi et al., 1983). The sampling sites of these rocks are plotted in Fig. 3.

Data for six gneissose granodiorites give an isochron of 693 ± 16 Ma with an initial ratio of 0.70450 ± 0.00029. Yanagi et al. (1983) considered that the gneissose granodiorite (sample Y1401) should be excluded from the others owing to a development of recrystallized albite blobs in microcline crystals. Data for five gneissose granodiorites except Y1401 give an isochron of

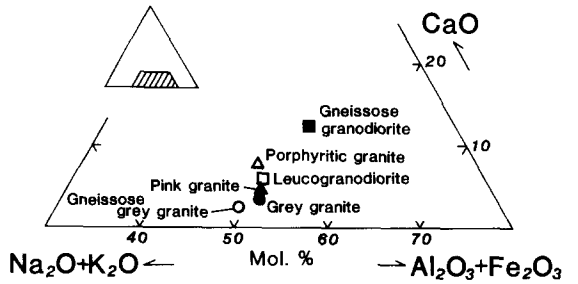


Fig. 6. Chemical compositions of the granitic rocks of the Seychelles plotted in the $CaO-(Na_2O+K_2O)-(Al_2O_3+Fe_2O_3)$ diagram (after Suwa et al., 1983, and Hoshino, 1986).

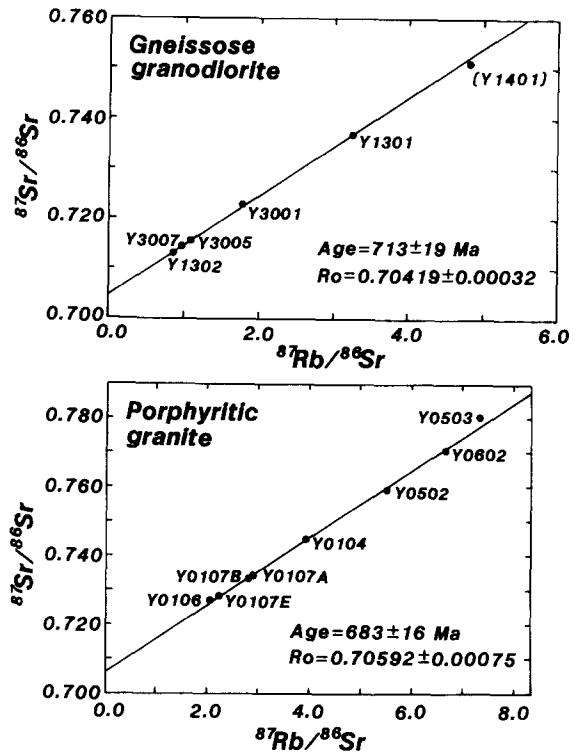


Fig. 7. Isochron plots for the gneissose granodiorite and porphyritic granite in the Mahé island group (after Yanagi et al., 1983).

713±19 Ma with an initial ratio of 0.70419±0.00032 (Fig. 7). This value of 713 Ma was recommended as the most reliable age for gneissose granodiorite (Yanagi et al., 1983).

Data for seven porphyritic granites give an isochron of 683±16 Ma with an initial ratio of 0.70592±0.00075 (Fig. 7).

Data for five grey granites give an apparent isochron of 570±5 Ma with an initial ratio of 0.7275±0.0012. Yanagi et al. (1983) considered that this apparent isochron age was erroneous due to the recrystallization of microcline.

A pair of whole-rock and biotite samples of the olivine bearing biotite–clinopyroxene diorite in the North Island gives an age of 60±4 Ma. The initial ratio of the diorite is very low at 0.70316±0.00010.

5. Remanent magnetization of Precambrian granitic rocks

5.1. Measurements of natural remanent magnetization (NRM)

408 oriented granite hand-specimens were collected for palaeomagnetic study from 49 sites in the Mahé island group (Fig. 3). Two or three core samples, 23 mm in diameter and 23 mm long, were drilled from each hand-specimen in the laboratory. Directions and intensities of remanent magnetization were measured on 122 granite specimens from 49 sites, using the spinner magnetometer having a noise level of 10^{-10} A m^2.

The NRM directions of the granitic rocks are found to be widely scattered as shown in Fig. 8a. The NRM intensities of the granitic rock are generally strong, of the order between 10^{-4} and 10^{-3} A m^2/kg.

5.2. Alternating field (AF) and thermal demagnetization

Three to five pilot specimens from each site (total 197 specimens) were subjected to the progressive AF demagnetization at 10 mT field increments, up to maximum peak field of 100 mT. Stable NRM directions of 121 granite specimens from 18 sites are grouped well after AF demagnetization of 30–80 mT (Figs. 8b, 9a and 9b). Intensities of the stable remanence are exceedingly reduced to 1/10–1/100 of the initial values (Fig. 9b). On the contrary, 76 granite specimens from 31 sites show no significant improvement of grouping in remanence direction after AF de-

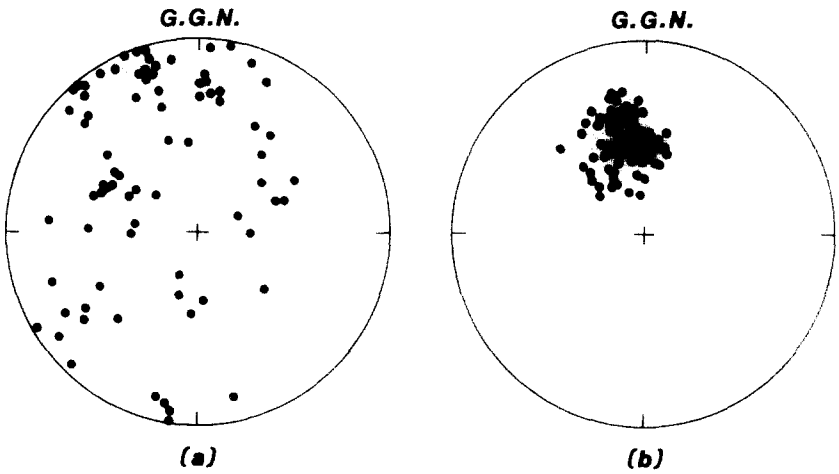


Fig. 8. Stereographic projection of NRM directions of well-grouped granite samples in the Mahé island group: (a) before cleaning, (b) after cleaning (Tokieda et al., 1983, and this study).

magnetization of 100 mT. Initial NRM intensities of the well-grouped specimens (around 10^{-4} A m^2/kg) are generally weaker than those of the non-grouped specimens (around 10^{-3} A m^2/kg).

A total of 30 specimens, both well-grouped and non-grouped specimens, were subjected to stepwise thermal demagnetization up to 560°C. Stable remanences are isolated at temperatures of more than 500°C. This indicates that the carrier mineral of the stable remanence is magnetite and, actually, magnetite is contained ubiquitously in all granite specimens. Directions of stable remanence of the well-grouped specimens are consistent with those obtained by AF demagnetization (Figs. 10a and 10b). On the other hand, the remanence directions of the non-grouped specimens are not grouped, even after the thermal treatment.

The results of the AF and thermal demagnetizations reveal that the isolated stable remanences of the gneissose granodiorite, grey granites exposed in the north of Mahé and pink granite at site 48 fall around a northerly mean direction with $D=348.5°$, $I=48.7°$ ($N=121$, $k=40$, $\alpha_{95}=2.0°$), as shown in Fig. 8b. The remainder, leucogranodiorite, porphyritic granites, pink granites except at site 48, grey granites exposed in the south of Mahé and gneissose grey granites show no grouping in their remanence directions. Thus, the non-grouping nature of the primary remanence directions is remarkable in the south of Mahé Island. This suggests that the primary remanence directions were disturbed by remanences of secondary origin, which phenomenon would have been caused by a tectono-thermal event some time after the emplacements of these granitic rocks, especially in the south of Mahé Island.

The Rb–Sr radiometric ages of the gneissose granodiorite and grey granite are 713 Ma and 570 Ma, respectively (Yanagi et al., 1983). However, as discussed in the previous section, 570 Ma for the grey granite is an apparent age. It would be petrologically reasonable to state that the grey granite emplacement did occur shortly after the porphyritic granite emplacement (683 Ma). Therefore, the age of primary magnetization of granitic rocks in Mahé Island is considered to be between 710 Ma and 680 Ma.

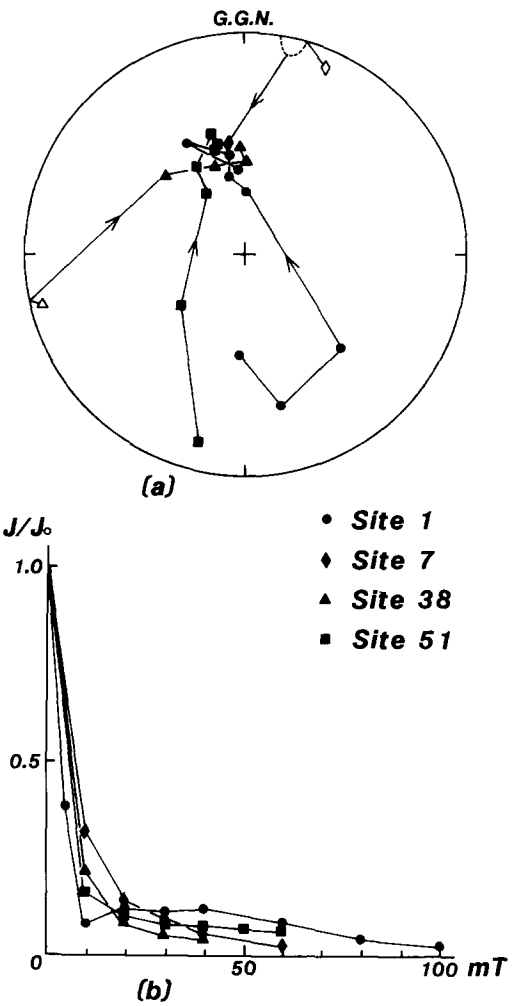


Fig. 9. AF demagnetization diagrams for representative granite specimens in the Mahé island group. (a) Stereographic projection showing progressive changes in remanence direction during AF treatment. Open symbols indicate negative inclination. (b) Progressive changes in remanence intensity during AF treatment.

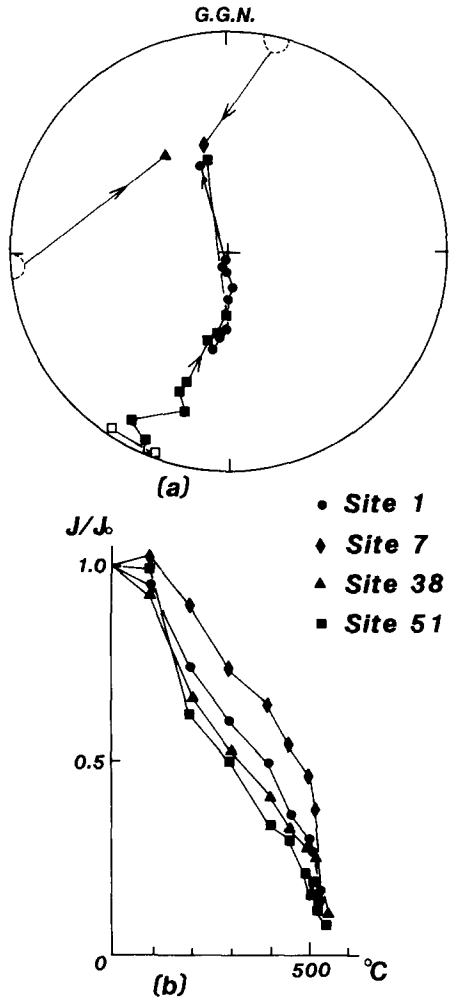


Fig. 10. Thermal demagnetization diagrams for representative granite specimens in the Mahé island group. (a) Stereographic projection showing progressive changes in remanence direction during thermal treatment. Open symbols indicate negative inclination. (b) Progressive changes in remanence intensity during thermal treatment.

6. Discussion

6.1. Palaeomagnetic reconstruction of the Seychelles

The stable remanences of the Precambrian granitic rocks of the Mahé island group yield a palaeomagnetic pole at 54.0°N, 38.2°E (Table 2). This pole position is almost identical with that obtained from a palaeomagnetic study on late Precambrian dolerite dykes of Mahé Island (Hargraves and Duncan, 1990). Hargraves and Duncan rotated the Seychelles first back to Madagascar and then back to Africa, in the order of sea-floor spreading events in the western Indian Ocean of which a model has been proposed by Emerick and Duncan (1982).

As discussed in the next section, we clarified an intimate relation in origin between the Seychelles and the Arabian–Nubian shield based on

Table 2
Summary of rotation of the Seychelles

Plate rotation pole	Rotation angle	Seychelles position	Magnetic pole position
		4.7°S, 55.5°E (present)	54.0°N, 38.2°E (V.G.P.)
18.7°N, 25.8°E (Seychelles–Madagascar)	10°CW	9.4°S, 51.3°E	51.1°N, 46.7°E
1.9°N, 105.6°E (Madagascar–Africa)	17°CW	5.1°N, 49.6°E	64.1°N, 63.5°E

petrological, geological, geochronological and geochemical evidence. Therefore, these data should also be taken into account in reconstructing the Seychelles.

We rotated the Seychelles in the same way as done by Hargraves and Duncan (1990), i.e. first back to Madagascar and then back to Africa as shown in Table 2. The Seychelles–Madagascar and Madagascar–Africa rotation poles are the same as those adopted by Hargraves and Duncan (1990), but the rotation angles are slightly modified so as to make the Seychelles' pre-drift position reasonable with regard to our petrological correlation. Consequently, the pre-drift position of the Seychelles is near the eastern end of the Horn of Africa (Fig. 11) and the restored magnetic pole is located closely to the N1 pole which was reported by Kröner et al. (1980) as a magnetic pole for the African continent in late Precambrian time.

A palaeomagnetic study by Embleton and McElhinny (1975) and a geological study by Rabinowitz et al. (1982) showed that Madagascar was located off the east coast of Africa adjacent to southern Somalia, Kenya and Tanzania before the break-up of Gondwanaland. It is noted that there is a good consistency in drifting direction between our reconstruction model for the Seychelles and their model for Madagascar, as shown in Fig. 11. It is interesting that the Seychelles in the estimated pre-drift position provide a link between the African, Arabian, and Indian plates in the Gondwanaland reconstruction by Smith and Hallam (1970).

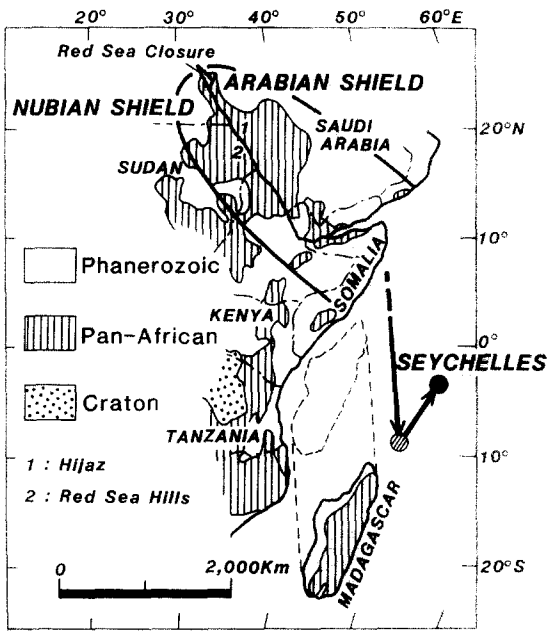


Fig. 11. Palaeomagnetic and petrological reconstruction of the Seychelles (reconstruction of Madagascar after Rabinowitz et al., 1982).

6.2. Petrological reconstruction of the Seychelles

The late Precambrian granitic rocks of the Seychelles are very similar to and are well correlated with those of the Arabian–Nubian shield as shown in the following.

6.2.1. Pan-African continental crust of the Arabian–Nubian shield

The Pan-African continental crust of the Arabian–Nubian shield evolved through a period of about 700 Ma (1200–500 Ma) by progressive

cratonization of numerous intra-oceanic island arcs. It is divided into four major divisions: Lower, Middle, Upper, and Post Pan-African. The plutonic calc-alkaline association in the Lower Pan-African (1000–1200 Ma) is gabbro-diorite–granodiorite. The plutonic calc-alkaline association in the Middle Pan-African (600–1000 Ma) is diorite–granodiorite. The plutonic calc-alkaline association in the Upper Pan-African (600–500 Ma) is granodiorite–granite. Finally, about 500 Ma ago, Post Pan-African alkaline and peralkaline magmatism with clear within-plate geochemical characteristics had replaced the previous calc-alkaline associations throughout the entire region (Gass, 1977, 1981; Engel et al., 1980; Bentor, 1985; Sylvester, 1989).

In the Seychelles, calc-alkaline granite was emplaced at 713 Ma, sub-alkaline granite at 683 Ma, and lastly alkaline granite at 570 Ma.

6.2.2. Central Hijaz region of the western Arabian shield

The late Precambrian granitic rocks occurring within a 44,000 km^2 area of the western Arabian shield are subdivided into older and younger assemblages (Jackson et al., 1984). The older plutonic assemblage (820–715 Ma) is related to a calcic, calc-alkaline suite of rocks ranging from gabbro through trondhjemite. The younger plutonic assemblage (686–517 Ma) is dominated by three coeval granitic associations: monzogranite, alkali granite, and alkali-feldspar granite associations. Monzogranites have I-type features, and alkali granites and alkali-feldspar granites have A-type features. The normative albite is more than normative orthoclase on both older and younger assemblages; it is especially clear in the older assemblages, Ab (40.0) $\gg$ Or (13.5); and it is clear in the younger assemblages, Ab (34.0) > Or (25.5).

In the Seychelles, calc-alkaline I-type granite (713 Ma) has normative Ab (44.3) $\gg$ Or (14.2), sub-alkaline A-type granite (683 Ma) has Ab (39.3) > Or (26.1), and alkaline A-type granite has Ab (34.6) > Or (28.5).

6.2.3. Red Sea Hills, Sudan

Granitic rocks in the Red Sea Hills are divided into an older diorite to granodiorite assemblage (815–724 Ma) and a younger alkali-feldspar granite assemblage (717–555 Ma) (Klemenic and Poole, 1988). The older and younger assemblages have a calc-alkaline character and low Sr initial ratios. Older granites are clearly plotted within the VAG (volcanic arc granite) field of Pearce et al. (1984). The VAG character of the younger granitic rocks is not as well defined as the older granitic rocks on the diagram. For the older granites, normative Ab (35.5) $\gg$ Or (7.4); for the younger granites, normative Ab (35.4) > Or (29.3); and for the youngest granites (569–555 Ma), normative Ab (32.6) < Or (35.0).

In the Seychelles, the Sr initial ratios of the calc-alkaline granite (713 Ma), the sub-alkaline granite (683 Ma), and the alkaline granite (570 Ma) are 0.70419, 0.70592, and 0.7275, respectively (Yanagi et al., 1983). In the Seychelles, the incompatible elements of the Mahé group granites are plotted in the boundary field between VAG and WPG (within-plate granite), and the Praslin group granites are plotted in the field of WPG (Richter, 1987).

7. Conclusion

From the above palaeomagnetic, petrological, geological, geochronological and geochemical data it is concluded that the late Precambrian granitic rocks of the Seychelles are very similar to and are well correlated with those of the Arabian–Nubian shield, and that the Seychelles first occupied the eastern end of the Horn of Africa and have drifted to the present position since the continental break-up as shown in Fig. 11.

Acknowledgements

We express our sincere thanks to Messrs. P. Shilton and G. Lionnet, Republic of the Seychelles for their most esteemed and invaluable assistance. We are very grateful to Professors T.

Yanagi, H. Umemura and M. Asami for their laborious collaboration in the field survey in 1981. The field work was made possible by the Grant-in-Aid for Overseas Scientific Survey from the Ministry of Education,Science and Culture, Japan, to which we express our thanks.

References

Agata, T. and Suwa, K., 1983. Opaque minerals from the Seychelles granitic rocks. 8th Prelim. Rept. Afr. Studies, Nagoya Univ., pp. 63–74.

Baker, B.H., 1963. Geology and mineral resources of the Seychelles Archipelago. Mem. Geol. Surv. Kenya, 3: 1–140.

Baker, B.H. and Miller, J.A., 1963. Geology and geochronology of the Seychelles Islands and structure of the floor of the Arabian sea. Nature, 199: 346–348.

Bentor, Y.K., 1985. The crustal evolution of the Arabo–Nubian Massif with special reference to the Sinai Peninsula. Precambrian Res., 28: 1–74.

Davies, D. and Francis, T.J.G., 1964. The crustal structure of the Seychelles Bank. Deep Sea Res., 11: 921–927.

Embleton, B.J.J. and McElhinny, M.W., 1975. The palaeoposition of Madagascar: palaeomagnetic evidence from the Isaro group. Earth Planet. Sci. Lett., 27: 329–341.

Emerick, C.M. and Duncan, R.A., 1982. Age progressive volcanism in the Comores Archipelago, western Indian Ocean and implications for Somali plate tectonics. Earth Planet. Sci. Lett., 60: 415–428.

Engel, A.E.J., Dixon, T.H. and Stern, R.J., 1980. Late Precambrian evolution of Afro–Arabian crust from ocean arc to craton. Geol. Soc. Am. Bull., Part I, 91: 699–706.

Francis, T.J.G., Davies, D. and Hill, M.N., 1966. Crustal structure between Kenya and the Seychelles. Philos. Trans. R. Soc. London, A, 259: 240–261.

Frankel, J.J. and Kent, L.E., 1964. On rocks from the Seychelles Islands. 22nd Int. Geol. Congr., Sec. 10: 161–190.

Gass, I.G., 1977. The evolution of the Pan African crystalline basement in NE Africa and Arabia. J. Geol. Soc. London, 134: 129–138.

Gass, I.G., 1981. Pan-African (Upper Proterozoic) plate tectonics of the Arabian–Nubian shield. In: A. Kröner (Editor), Precambrian Plate Tectonics. Elsevier, Amsterdam, pp. 387–405.

Hargraves, R.B. and Duncan, R.A., 1990. Radiometric age and paleomagnetic results from Seychelles dikes. Proc. ODP, Sci. Results, 115: 119–122.

Hoshino, M., 1986. Amphiboles and coexisting ferromagnesian silicates in granitic rocks in Mahé, Seychelles. Lithos, 19: 11–25.

Jackson, N.J., Walsh, J.N. and Pegram, E., 1984. Geology, geochemistry and petrogenesis of late Precambrian granitoids in the Central Hijaz Region of Arabian Shield. Contrib. Mineral. Petrol., 87: 205–219.

Klemenic, P.M. and Poole, S., 1988. The geology and geochemistry of Upper Proterozoic granitoids from the Red Sea Hills, Sudan. J. Geol. Soc. London, 145: 635–643.

Kröner, A., McWilliams, M.O., Germs, G.J.B., Reid, A.B. and Schalk, K.E.L., 1980. Paleomagnetism of late Precambrian to early Paleozoic mixtite-bearing formations in Namibia (south west Africa): the Nama Group and Blaubeker Formation. Am. J. Sci., 280: 942–968.

Laughton, A.S., Matthews, D.H. and Fisher, R.L., 1970. The structure of the Indian Ocean. In: A.E. Maxwell, M.N. Hill and W.H. Munk (Editors), The Sea. Wiley, New York, Vol. 4, Part 2, pp. 543–586.

Matthews, D.H. and Davies, D., 1966. Geophysical studies of the Seychelles Bank. Philos. Trans. R. Soc. London, A, 259: 227–239.

Matthews, D.H. and Reilly, T.A., 1964. Disappointing interim palaeomagnetic results from the Seychelles. Nature, 203: 1160.

Michot, J. and Deutsch, S., 1977. Les Seychelles, un nucléus sialique. Ann. Soc. Géol. Belg., 100: 147–156.

Miller, J.A. and Mudie, J.D., 1961. Potassium–argon age determinations on granite from the Island of Mahé in the Seychelles archipelago. Nature, 192: 1174–1175.

Pearce, J.A., Harris, N.B.W. and Tindle, A.G., 1984. Trace element discrimination diagrams for the tectonic interpretation of granitic rocks. J. Petrol., 25: 956–983.

Rabinowitz, P.D., Coffin, M.F. and Falvey, D., 1982. Salt diapirs bordering the continental margin of northern Kenya and southern Somalia. Science, 215: 663–665.

Richter, W., 1987. Die kontinentale Kruste im Bereich der Inselgruppe der Seychellen (Indischer Ozean). Mitt. Österr. Miner. Ges., 132: 39–59.

Smith, A.G. and Hallam, A., 1970. The fit of the southern continents. Nature, 225: 139–144.

Suwa, K., 1988. Travel through the Seychelles Islands. Bull. Nagoya Univ. Mus., 4: 139–165.

Suwa, K., Yanagi, T., Tokieda, K., Umemura, H., Asami, M. and Hoshino, M., 1983. Geology and petrology of the Seychelles Islands. 8th Prelim. Rept. Afr. Studies, Nagoya Univ., pp. 3–21.

Sylvester, P.J., 1989. Post-collisional alkaline granites. J. Geol., 97: 261–280.

Tokieda, K., Ito, H. and Suwa, K., 1983. Remanent magnetization of granites and dolerites from Mahé Island, Seychelles. 8th Prelim. Rept. Afr. Studies, Nagoya Univ., pp. 37–46.

Wasserburg, G.J., Craig, H., Menard, H.W., Engel, A.E.J. and Engel, C.G., 1963. Age and composition of a Bounty Islands granite and age of a Seychelles Islands granite. J. Geol., 71: 785–789.

Weis, D. and Deutsch, S., 1984. Nd and Pb isotope evidence from the Seychelles granites and their xenoliths: mantle origin with slight upper-crust interaction for alkaline anorogenic complexes. Isot. Geosci., 2: 13–35.

Yanagi, T., Wakisaka, Y. and Suwa, K., 1983. Rb–Sr whole rock age of granitic rocks from the Seychelles Islands. 8th Prelim. Rept. Afr. Studies, Nagoya Univ., pp. 23–36.